LESSON PLAN · SCRATCH JUNIOR (K–2) ADVANCED: GAME BUILDERS

Lesson 5: Mazes and Smart Paths

Module 2 | 30 minutes | Algorithms (planning paths) | KG–Grade 2 · ages 6–7

Objective: Plan a path through a maze, avoiding walls, and find a short way.

Success criteria

- I can plan a path through a maze.
- I can fix a bump.
- I can find a shorter way.

Key vocabulary

- **maze**: Paths with walls.
- **wall**: Something you can't go through.
- **shortcut**: A shorter way.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (30 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Chair maze (5 min): a child is guided through a chair maze by arrow cards.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	Solve a maze: learners follow the steps.
5 min	**Perform**	Cheese maze, Shortcut, Bump!
3 min	**Probe**	Gem maze
4 min	**Wrap-up**	Exit ticket: How do you fix a wall bump?

Unplugged warm-up

Chair maze (5 min): a child is guided through a chair maze by arrow cards.

Teaching the idea

Path planning in a constrained grid: trace, decompose into straight segments, encode with parameterised moves, debug collisions, optimise for fewer blocks.

Analogy: Finding your way through a hedge maze.

Live demo: Trace a maze path on the board; write one arrow card per straight part.

Questions to ask

Ask	Listen for
How many straight parts?	Count.
Where did it bump?	Find the block.
Is there a shorter way?	Try.

Differentiation

Support
Print the maze; let children draw the path first.

Stretch
Design a maze for a partner on squared paper.

Common misconceptions

Misconception	How to fix it
Walls can be walked through if you go fast	Walls always block.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

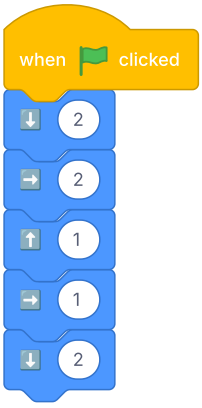
Exit ticket
How do you fix a wall bump?

Homework
Solve a maze in a puzzle book and count the turns.

Answer key — Kitabash activities

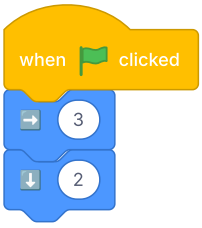
1. Cheese maze (perform · block code)

One solution:



2. Shortcut (perform · block code)

One solution:

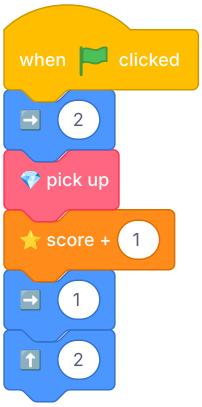


3. Bump! (perform · multiple choice)

Answer: Find the block where it bumped and change it

4. Gem maze (probe · block code)

One solution:



Answer key — worksheet

1. **Key words:** maze = Paths with walls. · wall = Something you can’t go through. · shortcut = A shorter way.
2. **Bump!:** A. Find the block where it bumped and change it — Fix the part that went wrong.
3. **Draw the path:** ⬇️2 ➡️2 ⬆️1 ➡️1 ⬇️2

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.